

All FAQs

FAQs About NIRAMAI Thermalytix

Are there any clinical publications showing the accuracy of Thermalytix? ?

Yes, Thermalytix has been clinically proven for high sensitivity and good specificity. There are 42 peer reviewed publications in top journals and conferences. Some clinical studies and real world evaluations published in top peer reviewed journal publications are listed below.

- **Study 1: BMJ Open Journal**, <https://pubmed.ncbi.nlm.nih.gov/34667011/>
(<https://pubmed.ncbi.nlm.nih.gov/34667011/>)
 - Multicentric study to evaluate the effectiveness of Thermalytix as compared with standard screening modalities in subjects who show possible symptoms of suspected breast cancer. *BMJ Open*. 2021;11(10):e052098
- **Study 2: ASCO JCO Journal**: <https://pubmed.ncbi.nlm.nih.gov/33001739/>
(<https://pubmed.ncbi.nlm.nih.gov/33001739/>)
 - Observational Study to Evaluate the Clinical Efficacy of Thermalytix for Detecting Breast Cancer in Symptomatic and Asymptomatic Women', *Journal of Clinical Oncology Global Oncology (JCO GO)* from *American Society of Clinical Oncology (ASCO)*. Oct 2020;6:1472-1480
- **Study 3: Frontiers in AI Journal**, <https://www.frontiersin.org/articles/10.3389/frai.2022.1050803/full>
(<https://www.frontiersin.org/articles/10.3389/frai.2022.1050803/full>)
 - A prospective evaluation of breast thermography enhanced by a novel machine learning technique for screening breast abnormalities in a general population of women presenting to a secondary care hospital. *Frontiers in Artificial Intelligence*. 2023 Jan 4;5:1050803
- **Study 4: International Journal Of Community Medicine And Public Health** <https://www.ijcmph.com/index.php/ijcmph/article/view/10481>

(<https://www.ijcmph.com/index.php/ijcmph/article/view/10481>)

- Feasibility and outcomes of using a novel artificial intelligence enhanced breast thermography technique, Thermalytix, in screening for breast abnormalities at primary health centres at the community level in South India, International Journal Of Community Medicine And Public Health, Vol. 9 No. 12 (Dec 2022)

- **Study 5: Lancet**

Oncology [https://www.thelancet.com/journals/lanonc/article/PIIS1470-2045\(22\)00419-3/fulltext](https://www.thelancet.com/journals/lanonc/article/PIIS1470-2045(22)00419-3/fulltext)

([https://www.thelancet.com/journals/lanonc/article/PIIS1470-2045\(22\)00419-3/fulltext](https://www.thelancet.com/journals/lanonc/article/PIIS1470-2045(22)00419-3/fulltext))

- Performance of artificial intelligence-based breast cancer screening in a community setting: a real-world evaluation study. The Lancet Oncology. 2022 Jul 1;23:S20

- **Study 6: SABCS**

2022 https://aacrjournals.org/cancerres/article/83/5_Supplement/P3-03-25/717325

- An Automated Risk Stratification System for Breast Cancer Screening using Thermalytix. Cancer Research. 2023 Mar 1;and San Antonio Breast Cancer Symposium Dec 2022:P3-03

You can access the full list of Peer-Reviewed Publications about Thermalytix **here.**
([/publications/](#))

Can you tell me about the screening test experience ?

?

What is the novelty in NIRAMAI solution?

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Is NIRAMAI Theramalytix different from Thermography?

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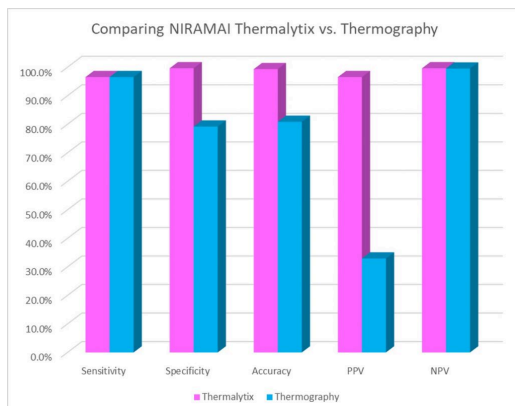
Yes, Thermalytix is different from Thermography. Thermography is visual analysis of thermal images, while Thermalytix uses sophisticated computer aided analysis using artificial intelligence for accurate analysis of thermal maps.

Thermography is an old technique that has been in practise by some medical professionals around the world for analyzing thermal images. It is recommended as an adjunct modality to detect early stage breast cancer. However, manual analysis of thermal images requires highly trained skilled professionals, and many times results in a number of false positives. NIRAMAI's Thermalytix solution fixes this accuracy issue using machine intelligence and provides an , automated method of quantfying thermal scans, and enabling screening for early stage breast cancer and other breast abnormalities. NIRAMAI Thermalytix has much higher accuracy than Thermography.

There are some important considerations showing why the Niramai solution is not the same as Thermography, and those differences are detailed below:

1. NIRAMAI Thermalytix© is different from Thermography. Though the sensing equipment used by Thermalytix is Infrared/Thermal Camera, similar to the one used by Thermographers, Thermalytix is based on analysis of temperature maps, where are thermography analyzes thermal images.
2. NIRAMAI Thermalytix Solution uses a unique AI-enabled software that gives objective assessment of the thermal analysis, which is missing in thermography. Additionally, demography data about the patient as well as a novel vascularity analysis is performed. So, we should treat Thermalytix as a new solution for detecting breast abnormalities.
3. Thermography clinics provide a report based on visual analysis of thermal images. Thermal images are very complex with about 400,000 temperature points shown as a color pixels, where each pixel can be any of the 2000 colors. Analyzing and interpreting these thermal images with just visual inspection is a huge cognitive overload for even an experienced thermographer. So, the manual diagnosis used in Thermography is error prone and hence the FDA guidance of using thermography as an adjunct to existing methodologies. However, NIRAMAI Thermalytix is not Thermography.
4. Thermography clinics are also NOT able to provide any report with one sitting of the patient. They usually ask the patient to come back in 3 months and do a visual inspection of 2 different sets of images taken 3 months apart. This results in potentially missing a case that was evident in the first sitting, and delaying the identification of malignancy for the patient. NIRAMAI solution is able to analyse the thermal images for every screening instance and provide a quantitative report.
5. NIRAMAI solution is based on machine learning models – that have the ability to analyse all the 400,000 color points both in imaging domain as well as temperature domain. A difference of 0.02 deg C is perceived by NIRAMAI software and hence the analysis is very sensitive, unlike thermography where two different shades of a color are hard to distinguish.

6. NIRAMAI machine learning models are calibrated with historical data labelled with current standard of care. The way NIRAMAI models were trained was based on labels learnt from mammography + sono-mammography + biopsy. It was not trained using reports of thermographers, hence NIRAMAI results are more accurate and better than current standard of care.
7. Niramai Thermalytix additionally uses patented vascular biomarkers which is able to detect very deep tumours, which are generally missed by thermographers, as there may not be an increase in temperature for the same.
8. In order to make an objective comparison between Thermography and Thermalytix, a clinical trial on 303 patients was conducted in a blinded retrospective analysis. This was peer-reviewed and published in International Conference on Advances in Breast Cancer Treatments, Kyoto, Japan, April 26-27, 2018. The results of this comparison showed a remarkable improvement in performance of Thermalytix over Thermography. An increase of 70% was seen in Positive Predictive Value over Thermography was seen, with very high levels of sensitivity and specificity. A copy of the paper that compares results of Thermography and NIRAMAI Thermalytix is available here (<http://waset.org/pdf/books/?id=89670&pageNumber=3>)



([https://niramai.com/wp-](https://niramai.com/wp-content/uploads/2019/12/WasetJapanResults.jpg)

[content/uploads/2019/12/WasetJapanResults.jpg](https://niramai.com/wp-content/uploads/2019/12/WasetJapanResults.jpg))

Thermalytix has regulatory clearance in India, Europe, UAE and Kenya at this time and usage in these countries does not fall under the purview of the above mentioned US FDA warning communication.

While Niramai has not yet applied for US FDA clearance for Thermalytix, Niramai obtained US FDA clearance (<https://niramai.com/niramai-bags-us-fda-clearance-for-their-medical-device-us-markets-now-open-for-the-innovative-indian-startup-niramai/>) for our SMILE-100 tool that has subset of features of Thermalytix.

What exactly is done in NIRAMAI breast screening? ?

Niramai solution is based on Thermography. Niramai thermal device which is placed 3 feet in front of the patient captures 5 thermal images of the region. Niramai software tool automatically analyses these images and sends the analysis report that is certified by a Senior Radiologist.

For more details, please watch this video <https://youtu.be/RT5njRmAdZk>
(<https://youtu.be/RT5njRmAdZk>)

What is the ideal time in a month to go for NIRAMAI/Thermography screening. ?

I am interested in going through NIRAMAI Breast Screening. Where should I come? ?

What is the minimum age for thermographic/NIRAMAI screening ?

Is Thermalytix painful like Mammography ?

Is there a regulatory approval for Thermalytix breast health screening ?

Yes. NIRAMAI Thermalytix has received regulatory approval from India, Kenya, UAE, Philippines and all countries in Europe (CE Mark). SMILE-100 (with subset of features of Thermalytix) is regulatory cleared by US FDA.

All Thermal devices used along with Thermalytix software are also US FDA cleared and have European CE Mark.

I am interested in buying NIRAMAI Solution. Whom should I contact? ?

If you are hospital or diagnostic centre interested in deploying this solution in your centre, please drop a note to contact@niramai.com. We will arrange a free demonstration of the solution and discussion with Niramai BD team. Niramai test is now deployed in 150+reputed hospitals and diagnostic centers such as Apollo Clinics, HCG, Narayana Health and more.

If you are a corporate interested in your employee wellness or an NGO interested in conducting screening camps for women, Niramai solution is available on monthly or daily-rental basis too. Please drop us a note on contact@niramai.com.

For deeper collaboration with Niramai, please write to partnerships@niramai.com

My relative has/had breast cancer. What is the risk of me getting breast cancer ?

How frequent do I need to have the screening ?

Things patient should do for perfect thermography/NIRAMAI screening before examination ?

When should I start going to breast cancer screening? ?

About Niramai

NIRAMAI Health Analytix is a Bangalore-based deep-tech startup addressing critical healthcare problems through automated solutions.

Our Mission

Our mission is to create a Universal Cancer Screening Method that can save lives.

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